Conformal campaigns: from coverage requests to audited consequences

Petrus H. Zwart*

Abstract

Conformal prediction returns prediction sets whose coverage is guaranteed, but in a screening campaign that guarantee is a request rather than a finding: the level is chosen and the threshold is built to deliver it, so every setting is equally guaranteed and coverage ranks none of them. It cannot say which setting to screen under, how many compounds will be assayed, or how large a round must be. We keep the two class-wise requests as coordinates instead and state what each one does as counts of the outcomes a campaign is planned on: confident hits, false leads, missed in-class compounds, compounds assayed. Expectations of those counts cannot be planned on or constrained with a stated probability, so we bound them in finite samples from a labelled audit set drawn from the library to be screened, and invert the bound to size a round. The audit is a second step of the same form, exempting the calibration set as the first exempts the training set, so its counts survive a calibration that does not match the screen; what it requires instead is the chemistry the batch will contain. In a DNA-encoded library screened for soluble epoxide hydrolase, an audit carrying the batch's own building blocks covered 0.97 ± 0.04 of the consequence lines while one matching only its proportion of new chemistry covered 0.63 ± 0.13, no better than ignoring the shift. Withholding blocks, the split used to guard against memorisation, cost 0.07 in ranking and two thirds of the coverage, so neither failure is visible in model-quality metrics. Whether the match holds is testable beforehand from structures alone, against a permutation " r"null. Dividing a pooled electrolyte compilation by the paper each measurement came from, " r"rather than at random, drops coverage from 0.97 ± 0.04 to 0.57 ± 0.14 at matched " r"prevalence, so provenance is part of what must be exchangeable.

1 Introduction

Conformal prediction has been advocated as a distribution-free form of uncertainty quantification.[1,2] It wraps any trained model, needs only a labelled calibration set, and returns for each new input a prediction set that contains the true label with probability at least $1-\alpha$ whenever calibration and future data are exchangeable. This guarantee, the coverage, requires no assumption about the model or the distribution of the data. That generality has made conformal prediction a standard tool for uncertainty quantification in machine learning and AI, and it is increasingly applied in chemistry and materials science.[3–5]

Here we apply conformal prediction to screening campaigns in chemistry and materials, in which a model decides which candidates go forward to a toxicity assay, a solubility or conductivity measurement, or synthesis. Increasingly such campaigns are run by automated and self-driving laboratories,[6,7] in which a model and a planner, rather than a person, decide what is made and measured. Because synthesis and characterization run in parallel, these campaigns proceed batch by batch: a model scores a candidate library and a rule decides what goes into the next plate or run, whose capacity is fixed. Unlike turn-by-turn optimization, a batch campaign often aims at covering a chemical or physical design space, mapping it or narrowing in on a set of optima (Section 4). The decision, whether made by a team or by a planner, is the same, and so is the gap we address. In a binary screen each compound receives one of four prediction sets: {out-class}, {in-class}, both labels, or neither. A team acts on these sets. It advances or discards compounds that receive a single label and sends the others for an assay.

To choose among these actions, a team needs to know what they will do: how many compounds will receive a single label, how many of those calls will be right, and how many compounds will be sent for testing. These consequences, counted in compounds, are the terms in which screening decisions are made and budgets and capacities are planned. Coverage, the native coordinate of conformal prediction, is a statement about the prediction sets; it informs a decision once it is linked to these consequences. That link has to be measured. A set containing both labels is always correct but sends the compound to an assay, so two classifiers calibrated to the same coverage can sort a batch very differently.

Predictive uncertainty is ordinarily reported for one prediction at a time, and its aleatoric and epistemic parts are defined for a single query instance;[8] reviews of uncertainty quantification for drug design and property prediction follow that convention.[9] Here the unit of prediction is the campaign rather than the compound. We report predictive uncertainty for the outcomes of a round, counted in compounds, and the same decomposition applies to it: the audit set reduces the uncertainty in the rates, and the size of the batch fixes the part that remains.

What we do and why. We give coverage a specific role. The coverage level is a setting the team chooses, separately for the in-class and out-class compounds, and changing it changes which compounds are advanced, discarded or assayed. We therefore treat the coverage request as the control: sweeping it moves the classifier through the space of possible decisions. A labelled *audit set*, a random sample of the library to be screened that was used neither for training nor for calibration, shows the consequences of each setting. From it we predict every screening outcome of a future batch, stated as counts of compounds, with prediction intervals that

are valid in finite samples and require only the exchangeability that coverage already assumes.

These intervals are the deliverable, and obtaining them is what the method is for. Before a batch is screened, the audit states how many confident hits, false leads, missed in-class compounds and measurements the batch will contain, each as a count, equivalently a rate per 1000, with an interval that holds with a specified probability. The intervals are realistic in both directions: they hold at the stated level, where the usual shortcuts are too narrow, and they are no wider than the size of the audit requires. A team can therefore plan on them: book the assays, budget the measurements and expect the hits.

Coverage is thus the coordinate for navigating decisions, and the audited consequences are what a team chooses by. Choosing uses an objective, such as the expected cost or yield of the screen. It usually also uses guarantees: a plan the team can rely on, in which, with a stated probability, no batch needs more assays than the laboratory or the robot can run and every batch delivers enough hits to proceed. Such guarantees are bounds on counts that hold with a stated probability. The coverage requests reduce the navigation of these many consequences to two coordinates, in which a team can explore scenarios, each a set of such guarantees, and see which settings satisfy them.

Section 2 sets out the outcome table, the threshold correction, the navigation by coverage and the audit intervals that are the method's output; Section 3 validates the intervals in simulation and on real data and applies them to solubility, toxicity and electrolyte screens; Section 4 gives a workflow and the limitations.

Contributions. The statistical tools are classical. What is new is the separation they make visible, and how they are combined to answer the questions a screening team asks.

1. *Coverage is a request, not a result; it is still a useful coordinate.* The coverage level is chosen and the threshold is constructed to deliver it, so every setting in the grid is equally guaranteed and the guarantee ranks none of them. What the requests do supply is a coordinate system. Every screening quantity is a sum of cells of a 2×4 table of prediction set by true label (Table 1), and with class probabilities the class-wise requests reach the two-threshold decision rules on the model score, up to the spacing of the calibration scores and the resolution of the grid, so a sweep traces the decisions available to the team in two coordinates. Each setting induces a decision geometry of the same form as the cost-optimal reject rule where the probability of being in-class increases with the score, and the audited consequences show which costs it serves. That condition is not universal and we report where it fails: across the twelve Tox21 endpoints the in-class fraction decreases over at least one adjacent pair of score bins in ten of them, significantly so for two (NR-AR and NR-ER-LBD), and for three endpoints the fraction is nearly flat over the lower nine bins with the signal confined to the highest (ESI Table S9). The containment argument and the decision-geometry reading do not apply to those endpoints.

2. *Two steps, two requirements, and only one of them is about the model.* Split conformal freezes a score fitted on training data and asks that calibration and future candidates be exchangeable. The audit repeats that move one level up: it prices a rule fixed by the training *and* calibration sets, and so exempts both. Coverage and predicted counts are therefore separate guarantees with separate requirements, and a campaign can hold either without the other. A rule calibrated on the wrong population, at the wrong prevalence, or against a cheaper label is a worse rule but not an invalid audit; an audit drawn from chemistry the batch does not share invalidates the counts however the rule was calibrated. We demonstrate both directions, and show that neither is visible in model-quality metrics: withholding building blocks cost 0.07 in ranking and two thirds of the coverage.

3. *Predicted consequences with specified certainty.* For each setting, finite-sample valid prediction intervals and one-sided bounds for the future counts of confident hits, missed in-class compounds, false leads and deferred compounds follow from the audit set.[10,11] Expectations of these counts cannot be planned on and cannot be constrained with a stated probability, which is what a bound supplies. With a Bonferroni correction over the sweep the bounds remain valid for whichever setting is chosen, so a setting can be chosen under chance constraints on assay capacity, budget or yield. We verify them in simulation and on three real datasets: toxicity (Tox21), aqueous solubility (AqSolDB) and electrolyte conductivity (CALiSol-23).

4. *Sizing a round.* Inverting the same intervals answers the question asked before a campaign starts: how many candidates a batch needs for a target number of hits to be reached with a stated probability. This builds on established estimates of campaign hit counts from calibrated probabilities[12] and extends them from an expectation to a bound. The distinction matters most where screens operate, since at low in-class rates the number of candidates required exceeds the expectation-based figure several-fold. The size of the audit enters that requirement directly, so audit measurements and synthesis trade against one another, and the trade speaks to the cost objection that led the expectation-based approach to forgo exchangeability in the first place. The exchange has a natural reading in the usual decomposition of predictive uncertainty:[8] the audit reduces the epistemic term, the uncertainty in the rates, while the size of the batch fixes the aleatoric term, which candidates happen to fall in it. The two are comparable when audit and batch are of similar size, so much larger audits buy little.

5. *Matching the audit to the screen, and checking the match.* Building the audit from the chemistry the round contains is therefore the design problem the guarantees reduce to. Where a round reaches into chemistry the audit does not hold, reproducing the batch's composition does not help and a pilot of that chemistry does; we give the pilot's required size. The requirement binds the calibration, audit and batch sets only, so a project may still separate its training compounds by scaffold or building block, and should. Whether the match has

been achieved can be tested before the batch is made, from structures and without measurements, because two samples from one distribution cannot be separated better than chance by a classifier held out of its own training data. The test is the classifier two-sample test,[13,14] read against a null obtained by permuting the two sets. When scores are class probabilities, the calibration set can stand in for the audit set, at a quantified cost in width.

Related work. Conformal prediction is established in cheminformatics, for toxicity, bioactivity, imbalanced data, applicability domain and drift.[3–5,15–21] The closest work estimates what a screening round will return. Buendia *et al.* predict the number of hits an iteration of an iterative high-throughput screen will retrieve, by summing per-compound calibrated probabilities from Venn–ABERS predictors over the selected compounds, and show across six targets that the sum tracks the realized count;[12,22] Svensson *et al.* link conformal predictions to the gain of a screen.[23] Estimating campaign-level counts is therefore established, and we build on it. A sum of calibrated probabilities is an expectation, and we add a probability statement about the realized count, which is what a plan that has to hold requires. Buendia *et al.* also identify the obstacle to such a statement: selecting the highest-ranked compounds leaves the calibration and screened sets progressively less exchangeable, retaining validity would require measuring a fresh random subset of the screened library at every iteration, and the cost of those measurements would make the approach impractical. They therefore keep the initial calibration set, and report that the realized hit count falls outside their expected limits in later iterations. That randomly chosen, measured subset is the audit set used here, and its cost can be set against the compounds it saves (Section 3.3). These studies report coverage and efficiency, or a gain, on test data; we add prediction intervals for what a future batch will contain. In statistics, the coverage of a conformal predictor given its calibration set is known exactly,[24–26] and related to tolerance regions and PAC prediction sets.[27,28] Risk-controlling methods[29–31] choose thresholds to bound one chosen risk. Conformal selection[32] selects candidates with false-discovery control, and set-valued classifiers with bounded class-wise error[33] minimize ambiguity. Decision-theoretic work links prediction sets to actions. Conformal decision theory calibrates a decision parameter to a target risk,[34] and Kiyani *et al.* showed that prediction sets are a sufficient statistic for risk-averse decision makers with a known utility, for whom acting on the worst case within the set is optimal.[35] The question for which costs a classifier, or an abstaining classifier, is optimal goes back to the reject option[36] and to ROC analysis.[37,38] Our aim is complementary. We assume no utility, report *all* screening outcomes of each coverage setting with valid intervals and bounds, and let a team choose afterwards under its own objective and constraints.

2 Methods

2.1 Setting and assumptions

We call the class with the property the screen looks for *in-class* (label 1) and the other *out-class* (label 0): toxic compounds in Tox21, soluble compounds in AqSolDB, and electrolytes that reach the conductivity target in CALiSol-23. A trained binary classifier is held fixed. It gives each compound a predicted probability of being in-class, $u = \hat{p}(\text{in-class} \mid x)$. A labelled *calibration set* sets the thresholds. A labelled *audit set*, which in machine-learning terms is a held-out test set, is used only to measure what the thresholds do; we call it an audit set throughout, because what it is for here is pricing a round rather than scoring a model. A *future batch* of m compounds will be screened. We assume throughout that the audit and future compounds are exchangeable with one another given the trained model and the calibration set, *i.e.* that their joint distribution is unchanged when they are reordered, and that the calibration set is independent of both. Where the coverage guarantee is also wanted, the calibration set must in addition be exchangeable with the batch; the two requirements are separated in Section 2.5 and they fail independently. What must be exchangeable is each candidate together with the label recorded for it, not the candidate alone, so the assumption can fail in two ways that look nothing alike: the compounds may be drawn from different populations, or the compounds may be interchangeable while the label recorded on one set is not the label recorded on another. The first is a question of which compounds a round contains and the second of what is measured on them, and both appear below. Exchangeability holds when all three sets are random samples of the library to be screened and carry the same label. The training set is deliberately absent from that list. The classifier is held fixed and enters only as a score, so how its training compounds were chosen bears on how good that score is and not on whether the intervals are valid. The two splits a campaign makes are therefore independent and answer different questions. Separating the training compounds from the rest by scaffold, series or building block guards against a model that has memorised its training chemistry, and lowers an optimistic estimate of how well it ranks compounds unlike the ones it was fitted on. Dividing what remains into calibration, audit and batch at random is what keeps the intervals valid. Both can be done at once, and should be: the arrangement that measures a model on unseen chemistry *and* prices the batch correctly is a split at the training boundary followed by random assignment within the deployment population. The unit of sampling is the candidate to be screened: a compound, or for electrolytes a formulation, not an individual measurement. It fails when the batch comes from other chemical series or scaffolds (Section 3.11), and again when the thresholds are set against a cheaper label than the one the campaign's claim concerns (Section 3.10). It fails too when the batch is chosen using the model, as in active learning, unless that selection is part of the design; we do not test that case, and methods for it are cited in the Discussion. The probabilities need not be calibrated in the usual sense; only their ordering matters.

2.2 Prediction sets and the outcome table

We use class-conditional (Mondrian) split conformal prediction,[1] which controls coverage separately for the in-class and out-class compounds. The label *out-class* is included in a compound's prediction set when $u \leq \tau_0$, where τ_0 is set from the out-class calibration compounds. The label *in-class* is included when $u \geq 1 - \tau_1$, where τ_1 is set from the in-class

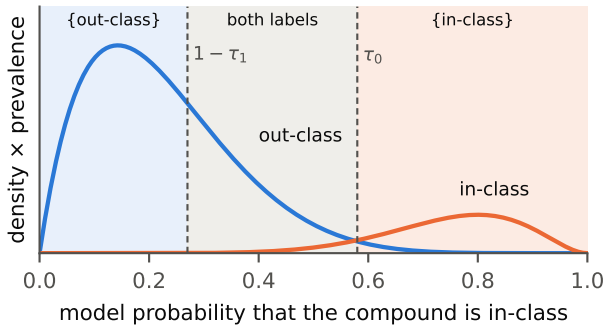


Figure 1: Class-conditional thresholds divide compounds into prediction sets. Curves show the distribution of the predicted probability of being in-class for out-class (blue) and in-class (orange) compounds, weighted by prevalence. Compounds left of $1-\tau_1$ receive {out-class}, those right of τ_0 receive {in-class}, and those in between receive both labels. If the thresholds cross, the middle band receives the empty set instead. The coverage request for each class moves one threshold.

Table 1: Outcomes of a binary conformal screen. Each compound falls into one cell; every screening quantity is a sum of cells.

True label	{out-class}	{in-class}	both	none
out-class	correct rejection	false lead	deferred	deferred
in-class	missed in-class	confident hit	deferred	deferred

calibration compounds (Fig. 1; ESI Section S1). Each compound then falls into one cell of a 2×4 table of true label by prediction set (Table 1). Every screening quantity is a sum of cells:

- missed in-class compounds are in-class compounds labelled {out-class};
- false leads are out-class compounds labelled {in-class}, *i.e.* compounds followed up that turn out out-class;
- deferred compounds receive both labels or none;
- coverage is the sum of the cells in which the true label is in the set;
- the hit rate of the confident calls is a ratio of two cells.

The cells in each row add up to the number of compounds of that class. Moving a threshold therefore moves compounds between cells of the same row, so the outcomes trade off against each other. With class probabilities, a compound can receive both labels or none, but not both kinds for the same thresholds (ESI Section S6). We report the two kinds separately, because an empty set signals a compound unlike the calibration data rather than an ambiguous one.[3]

2.3 Coverage requests and thresholds

The standard threshold guarantees coverage of at least $1-\alpha$ only on average over calibration sets. With tens of in-class calibration compounds, the coverage of the one model actually deployed can fall well below $1-\alpha$. The coverage observed over a batch of m_y future compounds of class y follows a Beta–Binomial distribution that depends only on the number of calibration compounds of that class and on the chosen order statistic. This result requires only exchangeability.[26] We choose the least conservative threshold for which batch coverage is at least $1-\alpha_y$ with probability at least $1-\delta_y$, separately for each class. A coverage request therefore has four numbers, $(\alpha_0, \delta_0, \alpha_1, \delta_1)$. The guarantee is stated for a planned number m_y of future compounds of each class, which we set from the batch size and the class proportions of the calibration set. That is a plug-in for a quantity the campaign does not know, and at low prevalence it is estimated from few compounds, so the tolerance δ is controlled for the planned count rather than the realized one and the deployed guarantee is approximate in it; the correction is conservative in the count, and the real-data checks below came out conservative, but we have not bounded the error. In an automated laboratory the batch is a plate or one run of the platform, so m is set by its capacity. The batch-free version,[24] which follows a Beta law, controls coverage only for an infinitely large batch (Section 3.1). As baselines we use the standard threshold and a one-sided correction based on the Dvoretzky–Kiefer–Wolfowitz inequality.[39]

Tied scores, for example from identical fingerprints, keep the coverage guarantee conservative, but they can break the count intervals that take a threshold from the same compounds they then count, because tied compounds cross it together; the audit intervals are unaffected, since the thresholds are fixed before the audit is counted. We therefore break ties at random: every compound receives an independent uniform number, and compounds are ordered by score and then by that number. This needs no knowledge of future scores and makes all the laws exact (ESI Section S2).

2.4 Navigating decisions with the coverage request

Coverage as a coordinate. With class probabilities, every rule that acts on single labels and defers the rest is a two-threshold rule on u: label out-class below one threshold, in-class above another, defer in between. The two coverage requests set the two thresholds, so a sweep of (α_0, α_1) over a grid, at fixed δ, traces the available two-threshold rules, up to the resolution of the calibration scores and of the grid. We keep one representative for each distinct pair of thresholds. Each setting maps to a vector of consequences, the counts of the outcome table and their bounds; the sweep traces a two-dimensional surface in this many-dimensional space, and its Pareto front over the outcomes that matter is the map on which a team explores its options.

The decision geometry of a setting. Without stated costs, a setting together with the usual rule (act on single labels, defer the rest) induces a decision geometry: two thresholds on the model score. Given only the prediction sets, this rule is the risk-averse (max–min) optimal use of the sets for any costs in which a deferral is cheaper than either error (ESI Section S7).[35] Its geometry matches Chow's cost-optimal reject rule[36] when the probability of being in-class increases with u, so each setting is expected-cost optimal for some region of cost ratios, possibly empty. Which costs a setting serves depends on the fractions of in-class compounds within each prediction set, which labelled data reveal; for a known utility, sets built for that utility can serve it better.[35] The coverage request therefore locates a decision, and its au-

dited consequences evaluate it.

Objectives and chance constraints. A team chooses a setting by an objective, such as the expected number of false leads, the expected cost or the expected yield, subject to constraints on what the campaign may consume or must deliver: at most U assays, at most U' missed in-class compounds, at least L confirmed in-class compounds in a batch of m. Each constraint must hold with probability at least $1-\varepsilon$, so it is checked against a one-sided bound from the audit set, not against the expected count. This is a chance-constrained program[40] with two decision variables, solved by enumerating the grid, in which the constraint probabilities are replaced by distribution-free bounds from the audit. A *scenario* is such a set of constraints. Its feasible settings form a region of the coverage plane, and the objective selects a setting within it. When the objective and every constraint are monotone in the outcomes (fewer measurements, false leads or misses, and more confirmed compounds, are never worse), an optimal setting always lies on the Pareto front of the audited outcomes, taken over the quantities the objective and constraints use: a dominated setting is never better than the setting that dominates it and meets no constraint that setting fails. For every such objective, the table of Pareto settings with their expected counts and bounds is therefore a sufficient summary, and no cost ratio is needed to decide. Only an objective that is not monotone, such as a target workload that should be neither exceeded nor undershot, needs the full map. Changing the scenario, for example tightening the assay budget, moves or shrinks the region and shows what the change costs. An empty region is itself a result: the requirements cannot be guaranteed with the labelled data in hand, and either a constraint must be relaxed or more compounds labelled.

2.5 Prediction intervals for the consequences

What the method returns. For a chosen setting and a planned batch of m candidates, the output is one line per outcome of Table 1: the expected count, and an interval that contains the realized count with a stated probability. Counts for a batch of 1000 are rates per 1000, so the same statement can be read either way. Tables 4 and 5 show the form. Everything else in this paper is built from these intervals: the sweep compares settings by them, the scenarios constrain them, and the audit-size rule plans their width.

From the audit set. Suppose an outcome (one cell, or a sum of cells) occurs K times among n audit compounds. Given the calibration set, the audit compounds and the m future compounds are exchangeable. Given the total count, the number in the audit set is therefore hypergeometric, whatever the data distribution. Inverting this exact hypergeometric test gives conservative limits $[L,U]$ for the future count K_m that hold with probability at least $1-\gamma$; they are quantiles of two Beta–Binomial distributions[10] (ESI Section S3). We use $\gamma = 0.05$ throughout. The construction holds for any outcome, any score and any fixed decision rule; it does not depend on conformal prediction. In the language of quality control the audit set is an acceptance sample, and the limits are the classical tolerance limits for the number of defectives in a future lot.[27,41]

Two steps, two requirements. It is worth naming what that conditioning does, because it accounts for the results below. Split conformal fits a score on training data, freezes it, and asks that calibration and future candidates be exchangeable; the training set is exempt precisely because it enters only through the frozen score. The audit repeats the move one level up. The rule it evaluates — the score together with the thresholds — is fixed by the training and calibration sets, and is then frozen, so the audit asks that audit and future candidates be exchangeable and exempts both of the sets that built the rule. From the audit's point of view the calibration set is training data. The argument is the same one twice, though not the same formula: a rank is uniform and a split of a count is hypergeometric for the one reason that under exchangeability every ordering, and so every subset of a given size, is equally likely.

Each step therefore carries its own guarantee and its own requirement, and a campaign can meet either without the other. Coverage is the first step's guarantee and needs the calibration set to be exchangeable with the batch. The predicted counts are the second step's and need the audit set to be, whatever the calibration was built from. So a rule calibrated on the wrong population, at the wrong prevalence or against a cheaper label is a worse rule but not an invalid audit, while an audit drawn from chemistry the batch does not share invalidates the counts however well the rule was calibrated; both appear below. The same reading explains why the calibration set cannot freely audit itself: for the second step it is training data, and a rule may not be priced on the data that set it (Section 3.11).

What the intervals cover. The intervals cover the realized count in the future batch; no fixed underlying rate is assumed. They are conservative because counts are discrete. Each interval is valid for one outcome and one setting fixed before the audit set is seen. Choosing a setting by looking at its audited outcomes and then quoting its intervals is optimistic. Intervals computed at level γ/J for each of the J requests in the grid hold simultaneously, and therefore remain valid for whichever setting is chosen. For a scenario with C bounded outcomes, each one-sided bound is computed at level $\varepsilon/(JC)$, so that every bound of every setting holds together with probability at least $1-\varepsilon$ and the chosen setting satisfies all constraints with at least that probability. Because the bounds hold jointly for all settings and all bounded outcomes, a team may pose any scenario over these outcomes, and change it, after looking at the map, without losing the guarantee. All intervals and bounds are conditional on the calibration set, that is, on the thresholds actually deployed.

What makes an interval useful. Validity alone is attained by the trivial interval $[0,m]$, which contains every possible count of an outcome in a batch of m and says nothing. An interval therefore earns its place by its width, and we report how many times narrower it is than that baseline, $m/(U-L)$, alongside coverage, following the principle of sharpness subject to calibration.[42] Width is set by the audit size and by how well the model ranks, and it can be planned before any labelling (Section 4). Whether the rule is better than no model at all is a separate question about the rule, not about a future count, and we test it on the audit set.

From the calibration set alone. With class probabilities and class-conditional thresholds, the calibration set can partly serve as its own audit. Whether an out-class compound receives the label *in-class* depends only on the threshold set from the in-class calibration compounds, so for this event the out-class calibration compounds are exchangeable with future out-class compounds and the audit construction applies to them. Whether an out-class compound *keeps* the label out-class is a coverage event, whose batch count follows the exact rank law used for the thresholds.[26] Each cell of a row is one of these two events or what remains after them, so computing both at level $1-\gamma/2$ bounds every cell of the row; a mixed batch adds an interval for the number of in-class compounds (ESI Section S4). Evaluating a class's own threshold on the compounds that set it, as is done when ordinary rates are computed on the calibration set, is not allowed; it makes the counts too narrow (ESI Section S5).

2.6 Experiments

Simulations. Eight designs are described in ESI Section S8:

1. audit-set intervals, with 10^4 replicates, each drawing its own data distribution;
2. finite libraries: 20 fixed libraries, each split at random 1000 times, which gives exchangeable data that are not independent;
3. threshold correction, with absolute Cauchy scores, $n =$ 50–500, $m = 100$ and 10^6 draws;
4. expected cost of the swept settings, computed exactly on the population, and selection: 4000 replicates of choosing among 100 settings under an assay capacity, with and without a Bonferroni correction;
5. scaffold dependence: compounds in clusters that share a shift of the predicted probability, with the future batch drawn from the calibration clusters or from new clusters;
6. tied scores: a cluster of identical scores at a threshold, with ties kept or broken at random;
7. calibration-only intervals: 3000 replicates per calibration size, compared with spending the same labelling budget half on calibration and half on an audit set.
8. a two-level draw, in which a rate is drawn and then the calibration, audit and future counts are drawn given it, used to separate a statement about the long-run rate from a statement about the realized count of one batch.

Data. Molecules were standardized with RDKit[43], which keeps the largest fragment, neutralizes it and writes canonical SMILES; duplicates were then merged. One model was used throughout: Morgan fingerprints[44] (radius 2, 2048 bits) and CatBoost gradient-boosted trees[45] (500 trees, depth 6, learning rate 0.1) with fixed hyperparameters. The experiments concern the coverage settings and their audited consequences for a given model, not the choice of model. For the twelve Tox21 endpoints,[46,47] each endpoint was split 50/25/25 into training, calibration and audit sets: 50 times at random and 20 times by Bemis–Murcko scaffold.[48] Under the scaffold splits we also drew pilot samples of 200, 500 and 1000 compounds at random from the audit scaffolds, used each as the calibration set, and screened the remaining compounds. For AqSolDB,[49] compounds with $\log S \geq -4$ were labelled soluble. Ten stratified 70/15/15 splits were made, the model was trained on all training compounds, and calibration and audit were restricted to $\log P > 3.5$. Solubility most often limits lipophilic compounds, so a screen of this subpopulation is a realistic use; it is also a subpopulation on which the model's scores are distributed differently from the training data. For CALiSol-23,[50] a compilation of measured conductivities of lithium-salt electrolytes, each formulation (salt, concentration and solvent fractions) was labelled good if its conductivity at 20–30 °C was at least 8 mS cm^{-1}. The same model was trained on all temperatures of the training formulations, with temperature, solvent fractions and fraction-weighted solvent descriptors as inputs, and scored each formulation at 25 °C. Four designs, 50 repeats each, drew calibration and audit sets from random formulations, from held-out salt–solvent systems, or, for a new salt, from the other salts or from a pilot sample of the new salt. To check the audit intervals on real data, each audit set was split at random, 20 times, into an audit half and a stand-in future batch. A random halving makes the two halves exchangeable by construction, so these checks verify that the construction and its implementation are right and cannot fail for any model, representation or label noise; they are not evidence that an audit prices a batch assembled some other way. The designs that test that are the block hold-outs and the declared round, where the batch is a distinct set of molecules. For the DNA-encoded library,[51] a sample of one million molecules was taken from the released table and a molecule was labelled in-class if it carried the library's own binding call for soluble epoxide hydrolase, a rate of 0.74%. Each molecule is built from one building block at each of three positions, so designs were defined over blocks rather than molecules: assignment at random, and assignment of whole blocks at one of the three positions, so that a batch molecule carries a block the training molecules do not. A declared round, 30% of it built on first-position blocks absent from training, was then prepared for in three ways that differ only in where the calibration and audit sets draw their novel chemistry. Counts for this library are for a batch of $m = 100\,000$. A further division of the electrolyte formulations by the paper each measurement came from, rather than at random, is described in ESI Section S10. Details, counts and seeds are in ESI Section S9, and for the library in ESI Section S11.

3 Results

3.1 Predicted counts hold at the stated certainty

The audit predicts the count of each outcome in a future batch, with an interval meant to hold with probability 95%. The checks in this section impose exchangeability by construction, through random splitting, and so establish that the construction is correct rather than that it survives a real change of population; the sections that put that to the test come later. In simulation, the realized count in a batch of 1000 compounds fell inside its interval in 95.7–98.9%

of replicates, for every outcome and every audit-set size (Fig. 2). The intervals were also tight: for wrong single-label calls (false leads plus missed in-class compounds) they spanned about ± 51 per 1000 with 100 audit compounds and ± 20 with 1000 (ESI Table S6). The same held when fixed libraries were split at random, a design that is exchangeable but not independent. Averaged over libraries, coverage was 96.1–98.9%, and no library fell below 94.7% (Monte Carlo standard error 0.7% per library; ESI Table S5). The usual plug-in interval, Binomial quantiles at the audited rate, treats that rate as if it were known; it covered only 44–75% of batches with 100 audit compounds and 84.5–93% with 1000, and fell to 23% for the worst library (ESI Tables S6 and S5). The opposite shortcut, m times the Clopper–Pearson interval for the rate, ignores the sampling variation of the future batch: it held with 100 audit compounds (96.3–98.6%) but fell to 94.0–97.6% with 300 and 86.6–94.3% with 1000. Each shortcut omits one of the two sources of variation, the uncertainty of the audited rate and the variation of the batch, and fails where the omitted one dominates (ESI Section S8); the audit interval accounts for both. Scaled Clopper–Pearson intervals are therefore adequate only when the audit set is much smaller than the batch, and not for small batches such as a single plate. Bayesian posterior predictive intervals with a uniform prior were 0.6–6.1% narrower and close to nominal in these designs, but carry no guarantee; in a two-level design they fell to 94.8% (ESI Table S7). That is a trade rather than a refutation: a fifth of a point of coverage against a few percent of width, against an exact interval that is itself conservative by one to four points in the same tables. The audit interval held at its nominal level throughout. Width is what distinguishes these intervals from the trivial $[0, m]$: for a batch of 1000 the 95% intervals of the four constrained outcomes at the chosen setting of each scenario screen below spanned 16–109 counts, 9 to 63 times narrower than that baseline, and 36–204 after the joint correction over the sweep, 5 to 28 times narrower.

On real data the audit intervals behaved the same way (Table 2). Splitting each Tox21 audit set into an audit half and a stand-in future batch, the intervals covered 95.6–99.0% of the realized counts, for every outcome. Per endpoint, coverage was 95.8–97.1% under random splits. It was 95.0–97.5% under scaffold splits, in which the model had been calibrated on other scaffolds: the audit described the consequences correctly because it came from the compounds being screened. On AqSolDB, over all 132–144 settings of the sweep and all outcomes, the pointwise intervals covered 96.1% of 216,960 realized counts. With the Bonferroni correction over the 144 requests, every interval of every setting covered its count together in 98.5% of 200 audit–future splits.

Thresholds for small calibration sets. With the standard threshold, coverage over a batch of 100 compounds fell below the requested 90% in 35–42% of calibration draws for $n = 50$ to 500, against a requested tolerance of $\delta = 10\%$; the batch correction brought this to 4.7–9.9%, matching theory to within Monte Carlo error (ESI Table S1). The batch-free correction is identical for $n \leq 100$ but too permissive beyond it, falling short in 11.8% of draws at $n = 150$ and 21.2% at $n = 500$, so the batch form should be used whenever batch performance matters; the one-sided DKW correction was valid but far more conservative. The same holds on real data: per-class audit coverage fell below 90% in 55 of 600 random Tox21 splits for out-class compounds and 45 for in-class compounds, against 56.2 and 42.3 expected from the exact law for the thresholds used (ESI Table S12), where the batch-free correction fell short in 17% and 13% of splits and the DKW correction in 0.2% and 1.0%.

Table 2: Real-data check of audit intervals (nominal 95%). Each Tox21 audit set (about 1750 compounds) was split 20 times at random into an audit half and a stand-in future batch of the same size; entries are the percentage of realized future counts inside the interval. Random: 600 splits, 12,000 checks per outcome. Scaffold: 240 splits calibrated on other scaffolds, 4800 checks per outcome.

Outcome	random	scaffold
correct rejection	96.2	95.6
false lead	95.9	96.2
deferred out-class	95.7	95.6
confident hit	96.6	96.5
missed in-class	99.0	99.0
deferred in-class	96.9	96.4
single label	95.8	95.8
deferred	95.8	95.8

3.2 What accepting the default setting costs

Table 3 compares, on the population, the expected cost of three choices: the best two-threshold rule without constraint, the best swept setting with coverage of at least 90% for both classes, and the default setting with coverage of exactly 90% for both classes. Thresholds are set at the exact quantiles, without a finite-sample correction, so the table isolates the effect of the coverage setting itself. The constrained optimum cost exactly as much as the unconstrained one in 46 of 60 settings and within half a percent in a 47th, and at most 47% more; where it cost more, the unconstrained optimum misclassified more than 10% of one class. The default setting, chosen without the costs in view, cost more. With a miss ten times as costly as a false lead, it cost up to 3.1 times the optimum for $\rho \geq 0.3$. When an assay was nearly free ($\rho = 0.1$), it cost up to 7.3 times the optimum at $\lambda = 10$, 13.6 times at $\lambda = 20$ and 32 times at $\lambda = 50$. The sweep of coverage settings thus contained the optimal rule in most cases, and navigating it with the costs in view finds that rule. That is a statement about the sweep, not a claim to beat a rule built for a known cost: where the costs *are* known and the probabilities can be recalibrated, a cost-optimal reject rule is the thing to use.[36] On the solubility screen the swept setting is indistinguishable from such a rule built on isotonically recalibrated probabilities, costing between 0.009 less and 0.013 more per compound across four cost regimes against standard errors of 0.004 to 0.038, and it is dearer by 0.011 to 0.171 than one built on Platt scaling (ESI Table S13). In the most lopsided regime, a miss fifty times a false lead and an assay half of one, both are beaten by deferring every compound: 0.665 per compound for the swept setting and 0.674 for the isotonic rule against 0.500 for deferring everything, which is a property of that cost structure rather than of either rule but is worth stating plainly. The case for the sweep is therefore that it needs no cost ratio and returns audited counts for whatever is chosen, not that it is cheaper when a ratio is in hand.

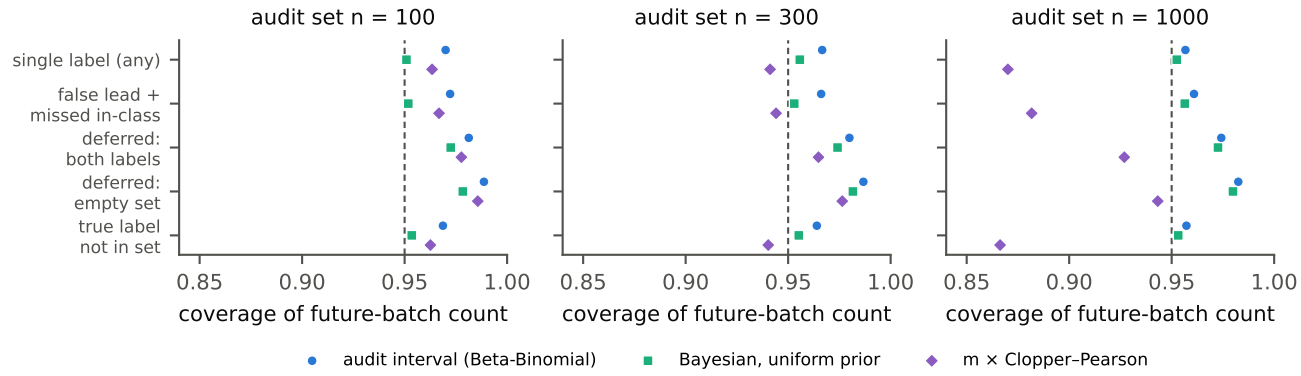


Figure 2: Coverage of 95% prediction intervals for the count of each outcome in a future batch of 1000 compounds, from an audit set of $n = 100$, 300 or 1000 compounds (panels); 10^4 replicates, each drawing its own data distribution. Audit interval: the Beta–Binomial limits of Section 2.5. Bayesian: posterior predictive with a uniform prior. $m\times$Clopper–Pearson: the interval for the rate scaled to the batch, which ignores the sampling variation of the batch. The plug-in Binomial interval, which ignores the uncertainty of the audited rate, covered only 44–93% and is omitted (ESI Table S6). Dashed line: nominal 95%. Monte Carlo standard error is about 0.2%.

Table 3: Expected cost relative to the best two-threshold rule (%), for three score distributions. λ: cost of a miss relative to a false lead; ρ: cost of an assay relative to a false lead. Constrained: best setting with population coverage $\geq 90\%$ for both classes. Default: population coverage exactly 90% for both classes. Exact population values, without the finite-sample correction. Full grid in ESI Table S3.

	$\rho = 0.3$		$\rho = 0.8$	
Scores, λ	constr.	default	constr.	default
well separated, 5	0	12	0	58
well separated, 20	0	3	0	9
overlapping, 5	0	31	6	5
overlapping, 20	0	100	0	16
balanced, 5	0	91	12	35
balanced, 20	0	453	9	203

Choosing the setting from audited outcomes needs the correction for selection. Among 100 swept settings, we chose the one with the fewest audited errors whose upper limit on deferrals fitted an assay capacity of 15% of the batch (ESI Section S8). With the Bonferroni correction over the settings, realized deferrals exceeded capacity in 0.03% of replicates; without it, in 0.47%. Both are below the nominal 5%, because neighbouring settings have strongly correlated counts, but only the corrected choice carries a guarantee. The correction cost 52.5 rather than 43.5 errors per 1000 compounds.

3.3 Sizing a round

A campaign asks the audit's question in reverse. A project needs a certain number of hits to continue, and what it must decide is how many candidates to put in the batch. The smallest batch whose lower bound on confident hits reaches the target answers that, and it follows from the same intervals by a search over the batch size. The interval is valid for each batch size fixed in advance, and the search makes the size a function of the audit count, so the probability statement does not transfer unchanged; because the lower bound increases with the batch size the selected size is the one at which the bound first reaches the target, which makes the reading natural but is not a proof, and we have not established a statement uniform in the batch size. The figures below should be read as the batch size the audited count implies, not as a size carrying its own guarantee.

The answer differs from the one an expectation gives, and the difference is largest where screens operate (Fig. 3, left). Dividing the target by the observed in-class rate, which is what a sum of calibrated probabilities amounts to,[12] asks for 1000 candidates to obtain 20 hits at a rate of 2%. The batch that delivers 20 hits with probability at least 0.95 holds 2056 candidates when the rate was measured on an audit of 384, and 1687 on an audit of 1000. At a rate of 50% the expectation asks for 40 and both audits give 52. The gap closes as the rate rises, because what the bound carries is the uncertainty in the rate itself, and that matters least when the rate is high and the audit has seen many in-class compounds.

The size of the audit enters the requirement directly, which prices audit measurements against synthesis. At a rate of 2%, an audit of 49 compounds leaves the required batch at 19 091 candidates and an audit of 384 at 2056, so 335 further audit measurements remove about 17 000 syntheses from the requirement; at a rate of 10% the same 335 measurements remove 201. Those are the requirements at the in-class count each audit is expected to see, and at these rates that count is small enough that the requirement is better read as a distribution. An audit of 49 at a rate of 2% sees one in-class compound with probability 0.37, giving 19 091, and none with probability 0.37, in which case no batch size reaches the target at all; two gives 2783. At an audit of 384 the count has a mode of 7 or 8 and the requirement runs from about 22 000 at two in-class compounds to 900 at fifteen, averaging near 3000 against the 2056 quoted at the expected count. The comparison between audit sizes is therefore a comparison of distributions: enlarging the audit from 49 to 384 removes the chance of an unanswerable round and roughly halves the typical requirement, which is the claim the figure supports, rather than a fixed saving of 17 000 syntheses. On this accounting the audit is not an overhead on the campaign but the measurement with the largest effect on what the campaign costs, and its effect is largest exactly where a screen's hit rate is low.

Three quantities are stacked in that figure, and separating them is what makes it planable. Were the rate known exactly,

the batch delivering 20 hits with probability 0.95 would still hold 1390 candidates rather than 1000, because which candidates fall in this particular batch is not something any amount of measurement can settle. Knowing the rate only from an audit of 384 raises the requirement to 2056, and knowing it from an audit of 49 raises it to 19 091. The first gap is the risk the campaign carries, the second and third are what ignorance of the rate costs on top of it, and only the latter two are for sale. A team that plans on the expectation omits all three.

How much of that is worth buying can be settled before any compound is measured, because the width of an interval has a closed form in the audit and batch sizes (ESI Section S3). Accepting a width r times the narrowest any audit could achieve requires $n = m/(r^2-1)$ audit candidates, a ratio that does not depend on the rate or on the batch size: an audit the size of the batch lands a factor $\sqrt{2}$ above that floor, one of $2.3m$ within 20% of it and one of $4.8m$ within 10%. The floor itself does depend on the rate, and at the rates a primary screen works at it is wide — for a batch of 1000 at a rate of 2% it is ±9 on an expected 20 — so a team should read the floor first and enlarge the batch, not the audit, when the floor is what binds. Both the floor and the pilot rule above are simple-random-sample results: where candidates fall in correlated series, each is inflated by the square root of the design effect, and enlarging a batch with analogues of series it already contains adds little to its effective size.[52,53]

The exchange has a limit (Fig. 3, right). An interval's width carries one term in the audit size and one in the batch size (Section 2.5): the first is the uncertainty in the rate, which the audit reduces, and the second is which candidates happen to fall in this particular batch, which it cannot. The two contribute equally when audit and batch are of the same size, and beyond that the batch term dominates, so an audit much larger than the batch buys little. For a batch of 50, an audit of 400 already leaves 89% of the interval's variance attributable to the batch.

Re-estimating between rounds costs little. A campaign that screens a first batch, revises the rate and sizes the next has looked at the data more than once, which a fixed-sample interval does not cover; spending the confidence level across a declared number of looks restores validity. Three looks rather than one raise the required batch from 309 to 350 candidates at a rate of 10%, and from 2056 to 2605 at 2%. The option to stop early, or to continue on a revised estimate, therefore costs between an eighth and a quarter of a batch.

3.4 What a measurement budget costs: a solubility screen

After standardization and de-duplication, 9254 AqSolDB compounds remained, 2467 of them with $\log P > 3.5$. In each of ten splits, the lipophilic calibration and audit sets held about 370 compounds each, 22% of them soluble ($\log S \geq -4$). Audit AUROC was 0.91–0.93 overall and 0.82–0.88 within the lipophilic subset.

The price of a budget. Cutting the measurement budget from 450 to 250 per 1000 lipophilic compounds cost 73 additional false leads, with all requirements guaranteed jointly at 95%, and no setting could be guaranteed below 200 measurements per 1000. At this audit size the guarantee itself

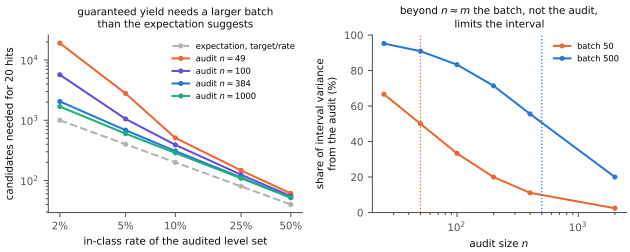


Figure 3: Sizing a round. Left: candidates needed to obtain 20 hits with probability 0.95, against the in-class rate of the audited level set, for four audit sizes; the dashed line divides the target by the rate, as an expectation does. Right: the share of an interval's variance contributed by the audit rather than by the batch, with the dotted lines marking where the audit matches the batch in size.

cost about 110 measurements per 1000: the chosen setting expects 286 and is bounded at 395. The rest of this section shows how these numbers follow from the map.

The decision space. Sweeping (α_0, α_1) over 12 values each, with $\delta = 0.10$, gave 132–144 distinct settings per split. Fig. 4 (left) shows their audited consequences for one split: one model, one endpoint, and outcomes ranging from 10 missed soluble compounds per 1000 with a third of the batch sent to measurement, to almost no measurements at the price of about 50 missed soluble compounds. Of the settings, 74–120 per split were Pareto-optimal for missed soluble compounds, false leads and measurements (ESI Table S10), so the front is the menu, and the team's constraints make the choice. Table 4 gives the audited consequences of the two extremes, with intervals for a batch of 1000. Setting A misses 10 soluble compounds per 1000 and sends 335 to measurement; setting B sends none to measurement and misses 52. Because A and B were picked by looking at the audit, the intervals that hold for them are the Bonferroni ones of Table 4, $[0,49]$ and $[15,114]$ missed, rather than the pointwise $[2,29]$ and $[29,84]$ that the same table reports for a setting fixed in advance. Over the ten splits the corresponding audited values were 0–20 missed and 203–522 deferred for setting A, and 26–61 missed and 0–42 deferred for setting B.

Scenarios. Two scenarios illustrate the choice (Fig. 4, centre and right). In both, a batch of 1000 lipophilic compounds must yield at least 60 confirmed soluble compounds and miss at most 90, and the objective is the fewest expected false leads. In the first, at most 450 compounds may be sent to measurement; in the second, at most 250. The limits were set after inspecting the map, which the joint bounds allow. Every constraint was checked against a one-sided audit bound, with the Bonferroni correction over the 144 requests and the four bounded outcomes, so that all constraints hold together with probability at least 95%. Under the first budget, 26 of the 144 requests were feasible; the chosen setting expects 115 false leads, 286 measurements and 23 missed soluble compounds per 1000, with bounds, holding together with probability at least 95%, of 395 measurements, 75 misses and at least 66 confirmed. Tightening the budget to 250 left 9 feasible requests and moved the choice to a setting expecting 188 false leads: the tighter budget costs 73 additional false leads per 1000. Choosing instead by the upper bound

on false leads, a risk-averse objective, selected the same settings in both scenarios. Both objectives are monotone, and as expected every chosen setting lay on the Pareto front of the four bounded outcomes, which held 105 of the 144 settings (ESI Section S7); the Pareto table is a complete menu for such objectives, and the constraints select from it. Over a range of budgets the price of the budget can be read off directly: the best feasible setting expected 221 false leads at 200 measurements per 1000, 188 at 250–300 and 133 at 350, and beyond 400 the budget no longer constrained the choice (115 false leads). Below 200 the feasible region was empty: no setting met all requirements with the labels in hand, so a team must relax a requirement or label more compounds. Against the trivial interval $[0, 1000]$, which validity alone attains, the chosen setting's 95% intervals were 9 to 26 times narrower for the four constrained outcomes, and 5 to 14 times narrower after the joint correction over the sweep. The advanced compounds, those given a single in-class label with no measurement spent, were 54% soluble against a prevalence of 22%, an enrichment of 2.4 times that is significant on the audit set after a Bonferroni correction over the 144 settings (one-sided Fisher exact test, $p = 2 \times 10^{-15}$), so the model demonstrably earned the measurements it saved. The gap between expected and bounded counts (286 against 395 measurements) is the price of the guarantee at this audit size, and labelling buys it down: at the same audited rate, half the audit (192 compounds) would give a bound of 432, twice the audit 372 and four times 358. A capacity rule chosen this way on one half of the audit set and checked on the other half never exceeded capacity in 200 splits (median upper bound 484, realized 304 measurements per 1000). No cost ratio is needed for this choice; the cost ratios for which each setting is expected-cost optimal are mapped in ESI Fig. S2.

3.5 One request, twelve sets of consequences: Tox21

After standardization and de-duplication, 7592 Tox21 compounds remained, with 5619–7017 labelled per endpoint and an in-class (toxic) prevalence of 2.9–15.9%. Here a confident hit is a compound confidently flagged as toxic, which a screen would set aside. Calibration sets contained a median of 44–228 in-class compounds per endpoint (ESI Table S12). Audit AUROC was 0.70–0.87 for random splits and 0.67–0.85 for scaffold splits. Before tie-breaking, a median of 3.3–4.3% of scores were tied, mostly from identical fingerprints.

The same request, very different consequences. All endpoints were calibrated with the same request, $\alpha = \delta = 0.10$ per class, using the batch correction. Yet the median fraction of compounds receiving a single label ranged from 13.6% (NR-PPAR-γ) to 62% (NR-AhR) (Fig. 5). Table 5 gives the audited consequences of the two extremes, for the split with the median single-label fraction. For NR-AhR, a batch of 1000 compounds yields 78 confident hits (audit interval 58–101) against 78 false leads (57–100). For NR-PPAR-γ, it yields 15 (6–26) against 89 (67–113), with 865 compounds deferred. Half of the toxic calls are correct for NR-AhR, one in seven for NR-PPAR-γ. The audit reveals these consequences before the screen is run. The fraction of in-class compounds among single-label "in-class" calls ranged from 13% to 58%, an enrichment of 3.0 to 6.5 times over prevalence. With the standard threshold the single-label fractions were larger, 28% and 72%, but that threshold's coverage of a class fell below the 90% target in 41.5% of the 600 endpoint–split pairs for in-class compounds and 47% for out-class compounds, and in 54–80% of splits per endpoint for at least one of the two.

3.6 Auditing a new salt: an electrolyte screen

CALiSol-23 holds 13,825 conductivity measurements of lithium-salt electrolytes from 27 publications.[50] After removing 523 rows without a conductivity and 23 rows from one source whose values lie about 100 times below all others for the same formulations (a probable unit error), 1495 formulations had been measured at room temperature, 275 of them (18%) at or above $8\,\mathrm{mS\,cm^{-1}}$. The threshold is close to the conductivity of standard carbonate electrolytes; 1 M $LiPF_6$ in EC/EMC (3:7) measures 9.4–9.5 $\mathrm{mS\,cm^{-1}}$ in this dataset, so a good formulation is one within reach of that standard. A threshold of $10\,\mathrm{mS\,cm^{-1}}$ would leave about 18 good formulations in a calibration set of 350, too few for the request to be attainable (Section 3.11). Measuring conductivity is cheap compared with building and cycling cells, so here a deferral means a measurement.

A new salt. The sharpest test holds out a whole salt. It is a weaker test than the name suggests: the salt held out is $LiPF_6$, which is the incumbent of commercial practice, supplies 606 of the 1495 room-temperature formulations, and anchors the class boundary, so what is tested is transfer to a well-represented salt rather than to novel chemistry; its share of good formulations, 25%, also differs from the salts calibrated on, so prevalence moves with the chemistry here as it does in the library screen. A genuinely novel salt is not available in this compilation, where several candidates contribute a single room-temperature formulation each. With a model trained and calibrated on the other salts and the 606 $LiPF_6$ formulations screened, coverage of the good class averaged 29% and fell below 90% in all 50 splits (Table 6; Fig. 6, right). The audit of the screened formulations showed what this meant for the screen: at the default setting it reported a median of 174 missed good electrolytes per 1000 formulations, about two-thirds of the good ones, and 59 confident hits. Calibrating instead on a random pilot of 40% of the $LiPF_6$ formulations, about 240 measurements, restored it for the good class (5 shortfalls against 3.3 expected) and left a residual excess for the poor class (8 against 4.0, one-sided Poisson-binomial $p = 0.04$), and the audit showed the gain: 10 missed good electrolytes and 159 confident hits per 1000, for 308 measurements. Formulations split at random over all salts met the request (2 and 4 shortfalls against 4.7 and 3.1 expected), and the sweep of coverage requests gave 108–132 distinct settings per split, trading measurements (0–230 per 1000) against false leads (0–150) and missed good electrolytes (0–30) (Fig. 6, left). Holding out whole salt–solvent systems also broke the coverage guarantee (20 and 22 shortfalls against 4.5 and 5.0).

The audit carries the guarantee. In all four designs, including the two in which the coverage guarantee failed, an

Table 4: Audited consequences of two settings of the lipophilic solubility screen (split 0), per 1000 compounds: realized count in the audit set (384 compounds), with the 95% prediction interval for a batch of 1000 and the interval with Bonferroni correction over the 144 requests of the sweep, which remains valid for one nominated outcome after choosing a setting by its audited outcomes. Holding all six lines of a chosen setting jointly would require the level divided by 144×6 instead; the scenario bounds quoted in the text are computed that way, over the four outcomes they constrain. A: the Pareto setting with fewest audited missed soluble compounds, ties broken by fewer deferrals, $(\alpha_0, \alpha_1) = (0.30, 0.075)$; B: the Pareto setting with fewest audited deferrals, ties broken by fewer misses, $(0.275, 0.25)$; $\delta = 0.10$.

	Setting A			Setting B		
	audit	95%	Bonferroni	audit	95%	Bonferroni
missed soluble	10	[2, 29]	[0, 49]	52	[29, 84]	[15, 114]
false lead	232	[183, 285]	[148, 330]	221	[174, 274]	[139, 319]
soluble confirmed	177	[134, 226]	[103, 269]	172	[129, 220]	[99, 263]
insoluble rejected	245	[195, 299]	[159, 345]	555	[495, 614]	[446, 660]
deferred, insoluble	299	[246, 356]	[206, 404]	0	[0, 11]	[0, 26]
deferred, soluble	36	[17, 64]	[7, 92]	0	[0, 11]	[0, 26]

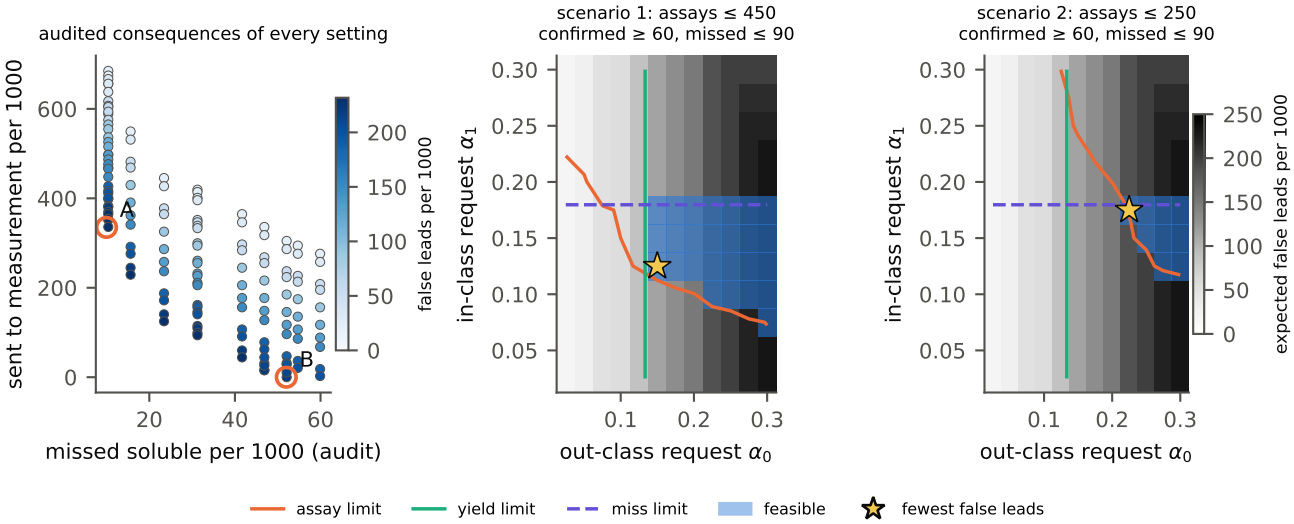


Figure 4: Navigating a solubility screen for lipophilic compounds (AqSolDB, $\log P > 3.5$; split 0) with one model. Left: each point is one setting of the coverage requests, placed by its audited missed soluble compounds and measurements, with false leads in colour; A and B are the settings of Table 4. Centre and right: two scenarios in the plane of the coverage requests, with the expected number of false leads (grey, the objective) for each request. Lines mark where the 95% joint bound of each constrained outcome equals its limit; blue cells satisfy all constraints with joint probability at least 95%, and the star is the feasible setting with the fewest expected false leads. The right panel tightens the assay budget from 450 to 250 per 1000.

Table 5: Audited consequences per 1000 compounds for the endpoints with the highest and lowest single-label fraction (batch correction), for the random split with the median single-label fraction (25th of 50). Entries: realized count in the audit set, with the 95% audit interval for a batch of 1000 in brackets. Calibration sets: 1385 out-class compounds and 197 in-class compounds (NR-AhR), 1526 and 37 (NR-PPAR-γ).

True label	Outcome	NR-AhR	NR-PPAR-γ
out-class	correct rejection	456 [417, 496]	32 [19, 48]
out-class	false lead	78 [57, 100]	89 [67, 113]
out-class	deferred	342 [305, 380]	850 [820, 877]
in-class	confident hit	78 [58, 101]	15 [6, 26]
in-class	missed in-class	7 [2, 15]	0 [0, 3]
in-class	deferred	39 [24, 55]	15 [6, 26]

audit set from the screened formulations reported the consequences correctly. Audit intervals for the default setting covered 95.5–100% of realized counts, pointwise intervals over the whole sweep 97.2–97.6%, and with the Bonferroni correction every setting and outcome was covered together in 98.4–99.2% of audit–future splits. A capacity rule chosen on one audit half, as a planner would choose it, never exceeded capacity on the other half, in 980–1000 checks per design. The intervals were also informative: for the scenario setting the 95% intervals were 12 to 63 times narrower than the trivial $[0, 1000]$, and 7 to 28 times narrower jointly over the sweep. The advanced formulations were 71% good against the 15% prevalence of the audit set of the split these maps use, which is lower than the 18% of the room-temperature compilation as a whole, an enrichment of 4.7 times (one-sided Fisher exact test on the audit set, $p = 2 \times 10^{-37}$, Bonferroni over 144 settings). Scenario maps for the electrolyte screen behave as for solubility (ESI Fig. S3). Intervals from the calibration set alone covered 99.7% of counts for random formulations; for a new salt they rely on a calibration that does not match, and a separate audit is the tool to use.

Per plate. A self-driving platform measures plates rather than batches of 1000. For the pilot-calibrated rule, an audit of 364 formulations gives, for one 96-well plate, 30 expected measurements with a 95% interval of 20–40; scaled Clopper–

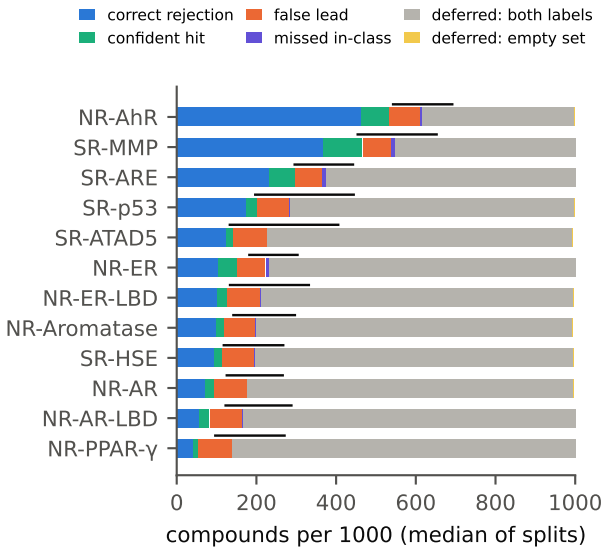


Figure 5: Twelve Tox21 endpoints calibrated to the same request ($\alpha = \delta = 0.10$ per class, batch correction) sort a screening batch very differently. Bars: median audited counts per 1000 compounds over 50 random splits, ordered by single-label fraction (medians need not sum to 1000); no empty sets occurred. Black lines: 5–95% range of the single-label count over splits.

Pearson would give 25–35, too narrow at this ratio of audit to batch size (Section 3.1). The one-sided bound of 38 measurements per plate holds with probability 95% for a single plate, and a bound of 43 holds for all ten plates of a campaign together (level 0.5% per plate).

3.7 An audit and a batch that are not two halves of one set

Every check of the audit intervals reported so far divides one labelled pool at random, which makes the two halves exchangeable by construction and establishes that the construction is right rather than that it survives a real separation. The electrolyte compilation lets the harder question be asked, because it records which paper each measurement came from. We therefore divided the formulations by source instead of at random, so that no paper contributes to both the audit and the batch, holding the model and the calibration set fixed across the divisions and halving nothing.

The intervals do not survive it. Over ten repeats, random halving covered 0.97 ± 0.04 of the consequence lines, as it must; partitioning the papers at random and letting each formulation follow its own paper covered 0.57 ± 0.14; and taking one paper's formulations as the batch and the rest as the audit, over the twelve papers large enough to serve, covered 0.65 ± 0.05. The median ratio of audit to batch in-class rate is 1.00, 0.98 and 0.90 in the three divisions, so this is not a change in prevalence; what moves is the relation between the score and the outcome, from one laboratory to another.

Two things follow. The first is a limit on this work's own evidence: the coverages quoted for solubility, toxicity and electrolytes under random splitting are verification, and the only divisions in this paper that test the intervals against a genuinely distinct batch — these, and the library's block holdouts — break them. The second is a limit on pooled compilations, which is where much of the data for materials screening comes from. An audit drawn from the library to be screened is not sufficient if the batch will be measured by someone else, on another instrument, with another calibration: provenance is part of what must be exchangeable, and a compilation of 25 papers is 25 populations. The instruction to audit the population you will screen therefore extends to the laboratory that will screen it.

3.8 Building the audit for the round it prices: a DNA-encoded library

The electrolyte screen showed that an audit drawn from the screened population bounds the consequences where a calibration set from elsewhere does not. A campaign poses a harder version of the same question, because the next round is often not drawn from the population already screened: it introduces chemistry that has not been made. What the audit must then contain is a question about its design, and a DNA-encoded library answers it cleanly, because the chemistry is built from a catalogue of building blocks and a round can be declared in those terms.

We use a sample of a published library[51] screened against soluble epoxide hydrolase, in which each molecule is assembled from one block at each of three positions and carries the library's own binding call; the in-class rate is 0.7%. At that rate a batch of 1000 contains about seven in-class molecules, too few for a setting confirming a handful of hits to be bounded away from zero, so counts in this section are given for the realized batch of each repeat, tens of thousands of candidates, rather than per 1000 as elsewhere.

How much is at stake can be measured in such a library by withholding blocks. Assigning molecules to training, calibration, audit and batch at random keeps all four exchangeable, and the intervals then covered 53 of 54 consequence lines over nine repeats, at their nominal level. Withholding instead every molecule carrying a given block at one position, so that the batch is built on chemistry absent from the calibration and audit sets, breaks them: over ten repeats each, coverage fell to 48% when the withheld position was the third, 37% when it was the second and 35% when it was the first. The six consequence lines of one repeat are cells of a single table on a single batch, so they are not independent checks and the repeat is the unit; taking it as such gives 0.48 ± 0.19, 0.37 ± 0.14 and 0.35 ± 0.12 against 0.98 ± 0.04 for random assignment, at ten repeats per design. Nothing in the model announces this. Its ranking of the batch stayed strong throughout, between 0.71 and 0.98 by area under the receiver operating characteristic curve, and the enrichment of its confident calls over the base rate was significant in every repeat of every design. The audit set does announce part of it, and we did not use that signal. In-class molecules are concentrated in a few blocks — five of the 306 at the first position carry 60% of the binders, one of them at a rate of 0.71 against the librarys 0.0074 — so partitioning whole blocks moves the in-class rate of the audit and of the batch independently, between 0.3% and 2.5% across these repeats. A team holding a labelled audit at 1.6% in a library whose base rate is 0.74% is told by that number alone that the two sets do not match, before any model is consulted. The coverage failures reported here are therefore a change in prevalence as well as a change in the mapping from score to rate, and the designs do not separate the two: the arm that holds, drawing the

Table 6: Electrolyte screen (CALiSol-23, conductivity $\geq 8\,\mathrm{mS\,cm^{-1}}$ at room temperature), 50 splits per design. Shortfalls: splits in which audit-set coverage of the class fell below 90%, observed/expected under the exact law. Audit coverage: percentage of realized counts in the stand-in future batch covered by 95% audit intervals, for the default setting (range over outcomes), for every setting of the sweep (pointwise), and for all settings and outcomes together with the Bonferroni correction over 144 requests.

		Shortfalls		Audit coverage (%)		
Calibration and audit drawn from	AUROC	poor	good	default	sweep	Bonferroni
random formulations	0.98	2/4.7	4/3.1	95.5–99.4	97.2	98.4
new salt–solvent systems	0.91	20/4.5	22/5.0	96.6–98.9	97.6	99.2
new salt, calibrated on others	0.92	0/4.5	50/1.9	95.9–100.0	97.3	98.9
new salt, pilot calibration	0.92	8/4.0	5/3.3	96.0–99.2	97.2	98.5

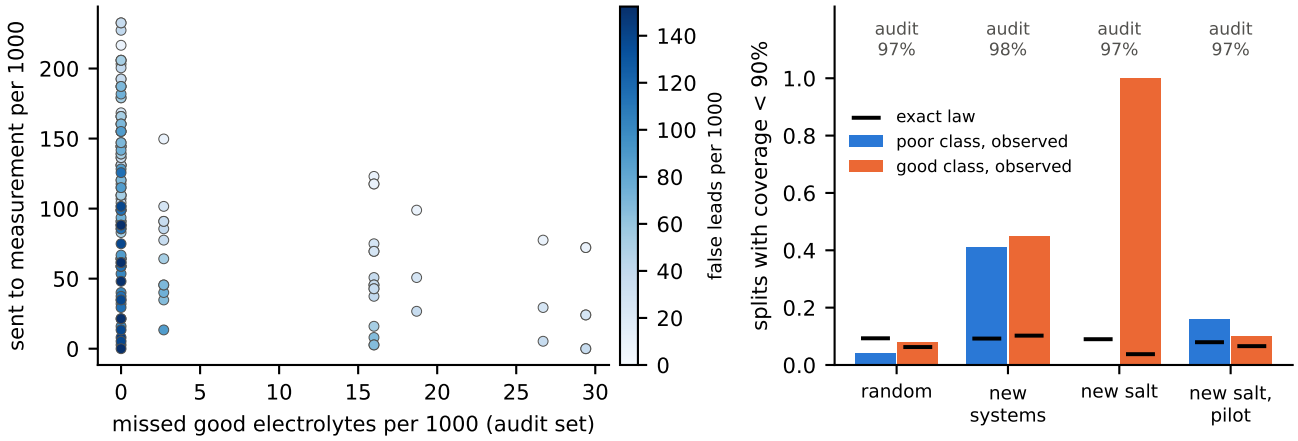


Figure 6: Electrolyte screen (CALiSol-23). Left: each point is one setting of the coverage requests for one random split, placed by its audited missed good electrolytes and measurements per 1000 formulations, with false leads in colour. Right: fraction of 50 splits in which coverage of each class fell below 90%, observed (bars) and expected under the exact law (black marks), for calibration on random formulations, on other salt–solvent systems, on other salts, and on a pilot sample of the new salt; above each design, the pointwise coverage of the audit intervals over the sweep.

audit from the batchs own blocks, matches the batchs prevalence by construction. Separating them would need block assignments stratified to equalize the two rates, which we have not run, and comparing the audits in-class rate against the librarys is a diagnostic a campaign should add whatever that comparison shows. In the worst of the ten position-1 repeats a batch of 42 145 candidates was promised between 578 and 723 confident hits, at a model AUROC of 0.89, and returned 119; per 1000 candidates that is an interval of 8 to 24 against a realized 3.

The modest size of that drop is itself the point, because withholding blocks is the kind of split a project adopts deliberately. Separating the held-out compounds by scaffold, series or building block is standard practice for checking that a model has learned chemistry rather than memorised its training set, and read as such the result here is reassuring: mean ranking fell only from 0.960 under random assignment to 0.892, 0.936 and 0.935 as the first, second and third positions were withheld, while over the same runs interval coverage fell from 98% to 35, 37 and 48%. The ranking figures are means of dispersed quantities — the position-1 repeats run from 0.708 to 0.976 against 0.950 to 0.968 for random assignment — and at a sub-percent in-class rate the area under the curve is governed by the ordering of the negatives and is deliberately blind to the prevalence changes these designs introduce, so it understates how little the ranking reveals. The two move together, the largest ranking cost accompanying the worst coverage, but they move on scales that are not comparable: a drop of 0.07 is one a project would accept as evidence that its model transfers, and it sits beside a guarantee that has lost two thirds of its coverage. Passing the check is therefore not evidence that the predicted counts transfer: it de-biases an estimate of how well the model orders compounds, and leaves untouched the mapping from score to rate, which is what a batch of new chemistry changes and what the intervals depend on.

A round is declared in which 30% of the batch is built on blocks at the first position that the model has never seen. Three ways of preparing for that round are compared, each supplying its own calibration and audit sets, all three sharing one model per repeat so that nothing differs between them but the sets themselves: calibration and audit drawn only from blocks the model trained on, as a team would do by habit; drawn with the same 30% share of new chemistry as the batch but from different new blocks, which is the position of a team that has measured some new chemistry and is about to make different new chemistry; and drawn with the same share from the batch's own new blocks, which is a pilot of the chemistry the round will contain.

Reproducing the batch's composition is worth nothing. Taking the repeat as the unit, as above, calibration and audit holding only familiar blocks covered 0.63 ± 0.13 of the consequence lines and the same sets carrying the batch's proportion of new chemistry from other blocks covered 0.63 ± 0.09: ten repeats cannot separate them, and neither comes near the nominal 95%. Sampling the batch's own blocks covered

0.97 ± 0.04, and was the only arrangement to cover all six lines in a majority of repeats, eight of ten against one and none. The pilot was also the sharpest of the three, its interval on confident hits spanning 70 candidates against 77 and 76, so the valid interval cost nothing in width here.

What distinguishes the three is therefore not how much new chemistry the audit contains but whether it contains *that* chemistry. The distinction matters because the first is cheap and can be arranged before synthesis, while the second requires measuring a sample of the new blocks, which is an experiment. Where a round introduces chemistry the audit does not hold, the guarantee is bought with that experiment or not at all.

3.9 Checking exchangeability before the batch is made

The preceding result is a design instruction, and a team following it would like to know whether it has succeeded. Exchangeability binds the calibration set, the audit set and the batch, and all three are sets of structures that exist before any of them is assayed, so the question can be put to the structures alone. We ask it as a two-sample problem: a classifier is trained to tell audit compounds from batch compounds using fingerprints and no labels. That is the pair the predicted counts depend on, since the audit prices whatever rule it is handed and asks only to be exchangeable with the batch; the calibration set may be compared with the batch in the same way when the coverage guarantee is also wanted. Under exchangeability the two are draws from one distribution, no classifier can separate them, and its area under the curve sits at one half.

Run across the four split designs, the test separated every arrangement that later failed and did not separate the one that held (Table 7); what it can refute is bounded by the representation, as the end of this section and the Limitations set out. It sits at 0.498 for random assignment, which covered 98% of consequence lines, and at 0.985, 0.989 and 0.997 for the three block holdouts, which covered 48, 37 and 35%; comparing the calibration set with the batch instead gives 0.501, 0.987, 0.990 and 0.999, so on these designs either pair would have raised the alarm. Beside it we report the share of batch molecules whose block at each position also occurs in the comparison set, which is the quantity the design controls directly. That share is one everywhere under random assignment and falls to zero at exactly the withheld position in each holdout, so the two statistics together give a verdict and name the axis responsible.

The test is one-sided and should be read that way. Separation is strong evidence against exchangeability, because two samples from one distribution cannot be told apart. Failure to separate is weak evidence for it, since a classifier may simply lack the features to see a difference that matters; the fingerprints used here cannot distinguish stereoisomers, for instance, whose in-class rates in this library differ severalfold. A screen that passes is therefore not certified, but a screen that fails is refuted, which is enough to be worth running: it costs no measurements, uses structures already known, and can be run before the plate is committed.

Table 7: A label-free check, run on structures alone before any compound is assayed, against the coverage each design later achieved. The domain classifier separates audit compounds from batch compounds, which is the pair the predicted counts require; one half means they cannot be told apart. The block shares are the fraction of batch molecules whose building block at each position also occurs in the audit set.

	domain	block share at position			
split design	classifier	1	2	3	coverage
random assignment	0.498	1.00	1.00	1.00	98%
withhold position 3	0.985	1.00	1.00	0.00	48%
withhold position 2	0.989	1.00	0.00	1.00	37%
withhold position 1	0.997	0.00	1.00	1.00	35%

3.10 Calibrating against a cheaper label

The audit sections asked which compounds a round must measure. The same assumption can fail along the other axis, in what is measured on them: the quantity a campaign cares about is an affinity, and what it can afford at scale is a proxy, such as a selection count or a single-shot readout. Thresholds are then set against the proxy while the statement the team wants concerns the affinity. The compounds may be a single random sample throughout, and they are in what follows, so nothing about the population is in question; it is the labelled pair that is no longer exchangeable, because the calibration set carries the proxy's label where the audit and the batch carry the campaign's. What that costs is not a question about how often the two labels agree.

One case is free, and for the reason that makes the construction rank-based. No part of the rule reads the numerical value of a score: a threshold is an order statistic of the calibration scores of one class, a prediction set follows from comparisons against it, and every count in the outcome table is a function of ranks. So the table is unchanged by any strictly increasing transformation of the score, which may therefore be a potency, a logit or an arbitrary monotone rescaling rather than a probability; we verify this on the library screen, where three such scales give tables agreeing cell for cell (ESI Section S12). The same argument applies to the label. If the proxy is an increasing function of the affinity, then being above a proxy cut-off and being above the corresponding affinity cut-off are the same event, so the two labels coincide and nothing in the analysis moves.

Real proxies are not increasing functions of anything. What governs the shortfall is local, and follows from the construction: the threshold sits at a fixed quantile of the score distribution conditioned on the *proxy's* label, whereas the coverage achieved is the score distribution conditioned on the *true* label evaluated at that same threshold. The shortfall is therefore a difference between two class-conditional score distributions at one point,

$$\text{achieved} - \text{nominal} = F(\tau \mid \text{true out-class}) - F(\tau \mid \text{proxy out-class}), \tag{1}$$

in which no term counts how many compounds the two labels disagree about. Disagreement costs coverage only insofar as it is structured with respect to the score.[1]

[1] Which is why a global summary of agreement, or a correlation between the proxy and the affinity, is the wrong thing to require of a proxy, and why comonotonicity is the right thing but is needed only locally: what Eq. 1 asks

Three carbonic anhydrase isoforms let this be separated, because selection counts and measured inhibition constants are published for the same targets[56,57] (Table 8). No molecule in that release carries both, so the two tables cannot be joined and no real selection count can be tested against an affinity here; the proxy is instead constructed from the affinities, which is what allows the structure of its errors to be set deliberately while the compounds and the score are held fixed. What follows is therefore a demonstration of the mechanism rather than a measurement of any particular assay's fitness as a proxy. Varying only the calibration label, we compare a strictly increasing relabelling of the affinity at matched prevalence, which is the control, against two proxies that relabel the same fraction of each class: one choosing those compounds uniformly at random, the other choosing the ones the score ranks most extremely against their class. The two differ from the truth on identically many compounds in each class, so their error rates, their ranking of the truth and the prevalence they imply are all equal; they differ only in where their mistakes sit in the ranking.

They behave nothing alike. At a fifth of each class relabelled, the random proxy left class-conditional coverage at 0.933 and 0.938 against a request of 0.90, falling short in 18 of 240 class-repeats, while the score-structured proxy gave 0.721 and 0.721 and fell short in 240 of 240, at every isoform and both cut-offs. Equation 1 accounts for this, and measuring both of its sides in the same run agrees to within 0.004 on average and 0.011 at worst across all 36 conditions: that is a consistency check on the accounting rather than a prediction, and its value is that it locates the quantity, not that it is accurate. The audit-set intervals, meanwhile, covered 95.2% of consequence lines over every condition, against their nominal 95%, and are flat across the table. They ask only that the audit and the batch be exchangeable, not that the rule be well calibrated, so they price whatever rule they are handed; a campaign that calibrates on a proxy keeps its predicted counts and loses its coverage guarantee, and only the audit tells it so.

That the requirement is local can be made exact, and in terms a campaign already uses. A team picks a quantile above which it will make compounds, and the rule here is of that shape, with one cut per class: at a request of $1-\alpha$ the out-class cut sits near the $1-\alpha$ quantile of the out-class scores and the in-class cut near the α quantile of the in-class scores, measured at 0.93 and 0.07 in these runs. Holding the relabelled share at a tenth of each class, so that the confusion matrix is the same in every condition, and moving only where in the within-class ranking those compounds are taken from, the damage follows the cuts and separates by class. Mistakes placed at the out-class cut displaced the out-class distributions by 0.036 and the in-class ones by 0.007; placed at the in-class cut the figures exchange, 0.007 and 0.036; placed anywhere between the two they left both at 0.007 to 0.009, which is the sampling floor. Coverage shortfalls track this, 54 of 120 class-repeats at a class's own cut against 9 to 19 elsewhere, and the audit intervals again covered 96.4% throughout. A proxy must therefore order compounds correctly in a narrow window around the quantile the campaign has chosen, and is free to misorder them everywhere else: the requirement attaches to a number the team states, not one it has to discover.

is that an isotonic regression[54,55] of the affinity on the proxy be increasing over the narrow score region where the thresholds fall, which the rule itself identifies as the region in which it declines to commit to a single label. Away from that region a proxy may misorder compounds freely. Fitting such a mapping needs compounds measured both ways, which a campaign can buy on a small sample.

Table 8: Calibrating against a proxy label on three carbonic anhydrase isoforms, at affinity cut-offs of 7 and 8, 20 repeats each. The two proxies relabel the same fraction of each class and so agree with the truth equally well by any label-only summary; they differ in whether their mistakes are placed at random or where the score ranks compounds most extremely. Coverage is class-conditional against a request of 0.90, pooled over isoforms and cut-offs; shortfalls count class-repeats below it.

	relabelled	agreement	coverage		audit
calibration label	share	with truth	out	in	intervals
increasing in affinity	0.00	1.000	0.924	0.923	96%
mistakes at random	0.05	0.950	0.929	0.933	95%
mistakes by score	0.05	0.950	0.873	0.872	95%
mistakes at random	0.10	0.900	0.932	0.936	96%
mistakes by score	0.10	0.900	0.822	0.824	94%
mistakes at random	0.20	0.800	0.933	0.938	95%
mistakes by score	0.20	0.800	0.721	0.721	95%

3.11 What the audit must sample

The audit describes a future batch drawn from the same library as the audit itself. In simulation, when compounds share scaffold-level shifts of their predicted probability, the intervals held when the audit sampled the scaffolds being screened: coverage was 95.4–97.9% (Fig. 7, left). Auditing other scaffolds does not substitute. When the future batch came from new scaffolds with a scaffold effect of one standard deviation on the logit scale, coverage fell to 71% for false leads, 83% for missed in-class compounds and 88% for deferrals; with 1.5 standard deviations it fell to 62–75%.

Tox21 scaffolds. On Tox21 scaffold splits, calibration on other scaffolds broke the coverage guarantee: per-class coverage fell short in 28% of splits for out-class compounds and 18% for in-class compounds, two to three times the exact-law expectation (22.8 and 18.2 of 240 splits). An audit set from the screened scaffolds still reported the consequences correctly (Table 2). Calibrating instead on a random pilot sample of the screened scaffolds restored the coverage guarantee: per-class violations then followed the exact law (ESI Table S8).

Solubility. We also calibrated on all calibration compounds (about 1390) and then screened lipophilic compounds. Coverage of the soluble class among lipophilic audit compounds averaged 72% and fell below the requested 90% in all ten splits. Calibrating on the lipophilic compounds only gave 94% (below 90% in one split of ten, as the request allows), at the price of more measurements: 462 rather than 209 per 1000.

How large a pilot must be. In the Tox21 pilots, violations for the in-class compounds were nonetheless frequent, 29%, 13% and 12% for pilots of 200, 500 and 1000 compounds, against 26%, 12% and 9% expected, because many pilots

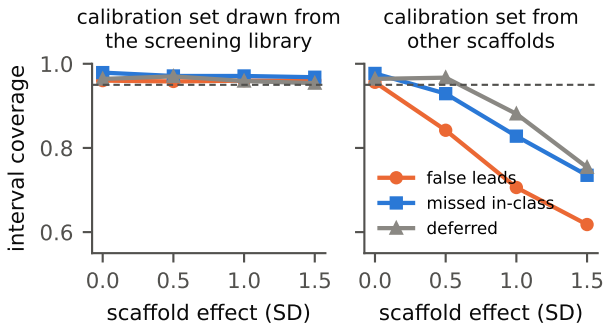


Figure 7: Coverage of 95% audit intervals when compounds are clustered in 60 scaffolds that share a shift of the predicted probability (3000 replicates per point). Left: calibration, audit and future compounds drawn from the same scaffolds, as with samples from the screening library. Right: future compounds from new scaffolds.

held too few in-class compounds for the request to be attainable. At $\alpha = \delta = 0.10$ at least 22–23 calibration compounds of a class are needed, whatever the batch size; 70%, 41% and 11% of the pilots had fewer (median 13, 29 and 54 in-class compounds). A pilot of $22/\pi$ random compounds at prevalence π has an expected in-class count of exactly 22, so it leaves the request unattainable in about half of draws; the pilots run here were infeasible in 70%, 41% and 11% of draws at 200, 500 and 1000 compounds. Sizing for a stated chance of feasibility rather than for the expectation, about $28/\pi$ reaches 22 in-class compounds nine times in ten, which is 980 for NR-PPAR-γ against the 770 the expectation asks for. At that minimum the request is met only by the most conservative threshold; each further labelled compound then buys single-label calls.

Using the calibration set as its own audit. When labels are too scarce for a separate audit, the calibration set can partly serve as its own, and intervals built that way covered the future count of every cell in 97.7–100% of simulation replicates, 98.7% of realized counts on Tox21 random splits and 99.7% on the electrolyte screen (ESI Section S4). They are not a substitute, for two reasons the paper has already given. At a fixed budget of 1000 labelled compounds they are wider in every cell than splitting that budget between calibration and an audit — 21–28% for false leads and missed in-class compounds, 31–39% for confident hits and correct rejections and about twice as wide for deferrals (ESI Table S2) — while the larger calibration set buys almost nothing in return, moving the quantile the thresholds use only from 0.944 to 0.940 for a batch of 96. And being the calibration set, they inherit whatever mismatch it carries: calibrating on other scaffolds cut their coverage to 74% and on other salts to 33%. The audit set is the only one of the two that can be drawn from the chemistry the batch will contain.

4 Discussion

The audit turns a conformal classifier into a forecast: before a batch is screened, it predicts how many hits, false leads, misses and measurements the batch will contain, with intervals that hold at a stated probability. The two class-wise coverage requests move a conformal classifier through the decisions available to a screening team, and for every setting the consequences are counts in one outcome table: compounds advanced, discarded and deferred, and how many of those calls are wrong. Linked to these counts by the audit, coverage becomes a coordinate a team can steer by. An objective, usually with guarantees on what a campaign will consume and deliver that hold with a stated probability, decides where to steer. The coverage plane is where such scenarios are explored, and an audit set from the library to be screened supplies the consequences, with finite-sample valid intervals that remain valid after the team has chosen a setting by looking at them. On Tox21, AqSolDB and CALiSol-23 the intervals held at their nominal level, also when the calibration came from other scaffolds or other salts, because the audit came from the screened compounds.

Using the method in a screen. The results suggest the following workflow:

1. Label a random sample of the library to be screened and use it as the calibration set. It must contain at least 22–23 compounds of the rarer class for a request at $\alpha = \delta = 0.10$ to be attainable, and more for a useful one: keep labelling randomly chosen compounds: $22/\pi$ at prevalence π makes the request attainable only about half the time, and about $28/\pi$ reaches it nine times in ten. Choosing compounds by their score to find the rare class faster breaks exchangeability unless the selection is accounted for.[58]

2. Label a second random sample as the audit set. Its size sets the width of the intervals: for an outcome with frequency p in a batch of m, n audit compounds give a 95% interval of roughly $\pm 2m\sqrt{p(1-p)(1/n+1/m)}$, about ± 35 per 1000 for $p = 0.1$ and $n = 400$ (ESI Section S3). Choose n from the width the decision needs, not from validity, which any audit size delivers. Read the floor first, since no audit narrows an interval below $\pm 2\sqrt{m\,p(1-p)}$: accepting r times that floor needs $n = m/(r^2-1)$, so an audit the size of the batch is the natural place to stop and an audit much larger than the batch buys almost nothing. Batch size decides what this costs. For a 96-well plate a few hundred audit compounds already sit within a fifth of the floor, and buying more is close to wasted; for a batch of 1000 the same audit sits more than twice the floor, and closing that gap means labelling on the scale of the batch itself. With class probabilities and few labels, the calibration set can serve instead, at a cost in width that Section 3.11 quantifies at the same labelling budget.

3. Draw the audit from the chemistry the next batch will contain. Where a batch introduces building blocks, scaffolds or series the audit does not hold, matching its composition is not enough: an audit carrying the same proportion of new chemistry drawn from different blocks performed no better than one ignoring the shift, while an audit sampling the batch's own blocks recovered the nominal level (Section 3.8). If part of the next batch is novel in this sense, a pilot of that chemistry is what restores the guarantee.

4. Before committing the batch, check that the calibration set, the audit set and the batch cannot be told apart.

Training a classifier to separate calibration from batch compounds on structures alone needs no measurements and can be run on the proposed batch: an area under the curve near one half is consistent with exchangeability, and anything well above it means the intervals do not apply (Section 3.9). Comparing how many of the batch's scaffolds, series or building blocks occur in the audit says which part of the chemistry is responsible.

5. Sweep the coverage requests and audit each setting: intervals for missed in-class compounds, false leads and deferrals in the planned batch, with a Bonferroni correction over the sweep.

6. State a scenario, an objective and the limits that must hold with probability at least $1-\varepsilon$ (assay capacity, budget, minimum yield), and choose the best setting among those whose audit bounds satisfy the limits. If the objective and limits are monotone in the outcomes, the table of Pareto settings is all the team needs to see.

7. Screen, and compare the realized counts with the predicted intervals as a check on exchangeability.

The audit decides which questions can be asked. Calibration and audit sets are usually treated as machinery, chosen for size and then forgotten. They are better understood as the statement of what the guarantees are about, because a sample licenses claims about the population it was drawn from and about nothing else. How the audit is built therefore fixes, before any model is fitted, which questions a campaign can answer, and three choices in building it each foreclose something. The population fixes the domain: an audit drawn from the top of a ranking prices further batches drawn from the top of that same ranking, and says nothing about compounds further down, which is where a campaign goes once the top is exhausted. The unit fixes what a count means: an audit sampled one compound per series estimates the rate at which series are real and prices a batch of new series, not a batch of analogues of a few confirmed ones, whose effective size is the number of series it contains. The chemistry fixes the reach: an audit holding only building blocks already screened prices batches built from those blocks, and carrying the same proportion of new chemistry as the batch without sampling that chemistry is not a substitute (Section 3.8). The label fixes the claim: what must be exchangeable is each candidate together with its recorded label, so a campaign that sets its thresholds against a cheap readout has made a statement about that readout and not about the quantity it cares about, however interchangeable the compounds are (Section 3.10). The score belongs to the same declaration, since thresholds and the sets they define move when the model is refitted, so an audit describes the deployment it was measured under.

Reading the four together makes the condition less forbidding than it first appears, because exchangeability is sufficient but more than is needed. A coverage request concerns one quantile, and what the request actually buys is an agreement between calibration and batch at the cut it induces; a discrepancy elsewhere in the ranking is free. That is why a proxy whose errors fall away from the cut costs nothing while one whose errors fall on it is ruinous at the same error rate, and it suggests that the familiar violations might be graded the same way, by where in the ranking a population differs rather than by whether it differs at all. We have tested this only on the label, where the structure of the discrepancy can be set deliberately; doing the same for a change of chemistry would ask how much of the loss under a block holdout is attributable to the neighbourhood of the thresholds, which the designs here do not separate.

Two consequences follow for practice. The question should be settled before the audit is built, because an audit assembled for one question usually cannot be repurposed for another; a campaign that discovers halfway through that it needs to speak about a different population has to measure again. And because an audit is also an option on the questions a campaign may ask later, it is worth spending part of it on compounds a project would not otherwise measure, lower-ranked compounds or a few members of each of several series, so that when the obvious batch is exhausted the next question can still be answered. Diversity in an audit is not a statistical nicety but the purchase of future questions.

Campaigns run batch by batch. Automated and self-driving laboratories synthesize and characterize in parallel, so their campaigns are sequences of batches, each with a fixed capacity, and the framework applies to each batch as it stands. Batch campaigns also tend to pursue different goals from turn-by-turn Bayesian optimization, which proposes one experiment at a time in search of a single optimum. Because a batch is evaluated in parallel, its members are chosen to be diverse,[59–61] and the campaign often seeks coverage of a chemical or physical design space: mapping a landscape, as in space-filling designs and closed-loop phase mapping,[62,63] contracting onto promising regions,[64] or finding a set of optima such as a Pareto front.[65] In each of these modes a batch is a selection from a scored library under a fixed capacity, and its consequences are what the audit predicts. The batch size m is the plate or run size, and the scenario constraints are the capacity of the platform, the reagent or instrument budget per run and the minimum yield needed to proceed. What the outcome table accounts for is which prediction set a candidate receives, not how the batch it belongs to was assembled. A planner that fills a plate with the highest-ranked candidates, or by a diversity-penalised acquisition, does not draw a random sample of the library, so the audit intervals do not price that plate; what they price is a random slice of it. A campaign that adapts therefore needs a freshly random audit slice in every round, and the guarantee attaches to that slice. Methods for data chosen by the model are the relevant prior art here rather than a complementary aside.[58,66,67] When the model is retrained between batches, the thresholds change, so the calibration, the audit and the map are redone for the new deployment. Two practices follow. First, reserving a small random audit slice in every batch, measured whatever the rule decided, checks continually that the screened library still matches the calibration data; it is the acceptance sample of quality control applied to each lot. Second, the guarantee holds per batch with probability $1-\varepsilon$; a statement over all K batches of a campaign needs level ε/K for each. Because the audit bounds are stated in counts, they can be checked automatically, and the Pareto table is a compact menu from which a planner, including an AI agent, can choose a setting without a cost ratio being stated; we do not test such a planner here.

Why parallel changes the economics. Automated campaigns are often pictured as sequential loops that adapt after every experiment, and a loop of that kind corrects its own errors: it proposes, measures, and the surprise moves the model, whether or not the uncertainty it reported was calibrated. Laboratories do not work that way, because the cost of an experiment is dominated by the round rather than by the candidate. A plate, a synthesis run or an instrument cycle costs nearly the same whether it is full or half empty, so cost rises in steps at each new round and barely at all within one. Three consequences follow, and they run against the intuition that smaller, more frequent batches are the way to keep a campaign honest.

The first is that an audit is cheap. When the round is the unit of cost, reserving part of a committed batch for a random sample of the library costs a few candidate slots rather than an extra round, so the measurements that price the next batch are close to free in money and are what the guarantees are bought with. The second is that correction is bought by composition rather than by cadence. Running more rounds to adapt sooner pays the fixed cost repeatedly, whereas reserving slots and spending the confidence level across a declared number of looks costs a fraction of one batch (Section 3.3). The third runs opposite to the sequential picture: the larger the batch that must be committed before anything is learned, the less opportunity there is to adapt within it, and the more of the decision rests on what was predicted beforehand. Parallelism and predicted consequences are therefore complements rather than alternatives. A laboratory that could run one cheap experiment at a time would have little use for an interval on a batch; a laboratory that must commit several hundred wells at once has the most use for one. On that reading a committed batch is an allocation under fixed capacity, divided between the candidates a setting advances, the random sample that prices the following round, and, where the next round will reach into chemistry the audit does not hold, a pilot of that chemistry (Section 3.8).

Limitations. All guarantees rest on exchangeability between the calibration, audit and screened candidates, each taken together with its recorded label. A change of chemical series, scaffold or assay breaks it, as the scaffold results show, and so does calibrating against a label other than the one the claim concerns, which breaks it without touching the composition of any set. So does a batch selected with the model, as in active learning, unless the selection is part of the design, and so does a library proposed by a generative model or extended into new chemistry, as the new-salt screen shows;[58,67] methods for shifted data are complementary. Compounds that share scaffolds are correlated, so the effective size of a pilot or audit sample is closer to the number of series sampled than to the number of compounds, as the account of what an audit decides sets out above. We can now put numbers to it. Decomposing the variance of the label within and between clusters gives an intracluster correlation of 0.63 for the electrolytes at the level of the salt–solvent system, with 5.8 formulations per system, hence a design effect of 4.1: the 1523 room-temperature formulations are worth about 375 independent ones. For the library at the level of the first-position block the correlation is 0.35 with 680 molecules per block, a design effect of 238, so the 200 000 molecules analysed are worth about 840. Clustering of this kind does not break exchangeability under random assignment, since which member of a cluster falls on which side is itself random, but it does mean the precision of an audit, the width of its intervals and the reach of a pilot are set by the number of clusters sampled rather than the number of candidates. The floor on interval width and the pilot-size rule given earlier are simple-random-sample results and should each be inflated by the square root of the design effect.[52,53] The pools used here are diverse enough for this to be immaterial, with 328 similarity clusters among 384 audited solubility compounds and over 1200 among the Tox21 compounds. It governs libraries built combinatorially or around a single target, and the DNA-encoded library is one: its random-assignment arm, which we report as the design under which the intervals hold, draws calibration, audit and batch from molecules that differ by single building blocks, so its effective sample size is smaller than its molecule count and the agreement we find there should be read as a check on the construction rather than as a measure of how much such a sample is worth. A structural check does not substitute for a pilot. We found no relation between how well an interval covered and how similar the held-out building blocks were to those the audit held; blocks indistinguishable by fingerprint gave the lowest coverage of all, being stereoisomers whose in-class rates differ severalfold, so similarity measured on an achiral representation cannot certify a batch that differs stereochemically from its audit. The same limit bounds the pre-flight test, which inherits whatever the representation cannot see: it separated every block holdout here, but a shift carried by features the fingerprints omit would pass it. Its verdicts are therefore asymmetric, refuting an arrangement rather than certifying one, and a richer representation would widen what it can refute without ever making a pass conclusive. Each interval covers one outcome of one setting; statements about many outcomes or settings need the Bonferroni correction, which widens intervals; with 144 requests and four bounded outcomes, the joint guarantee cost about 110 measurements per 1000 at an audit of 384 compounds. Larger audits lower this price, and bounds that exploit the monotone structure of the sweep could lower it further. The sweep reaches every two-threshold rule only with class probabilities, and the cost-optimal rule is a two-threshold rule only if the probability of being in-class increases with the score. Cost ratios are rarely known precisely, so we report the range of ratios for which a setting is cheapest. AqSolDB combines measurements from many sources, with a typical spread of 0.5–0.7 log units between sources;[49] labels near the cut-off at $\log S = -4$ are therefore noisy. CALiSol-23 likewise pools 27 sources, and several formulations were measured at many temperatures or concentrations, so neighbouring formulations are correlated. Its label also carries the provenance of the measurement rather than only the chemistry: 92% of room-temperature formulations have a single measurement in the 293–303 K window, and the temperature of that measurement is bimodal, 380 below 296 K and 942 above 300 K against 201 near 298 K, while conductivity rises by a median $0.136\,\mathrm{mS\,cm^{-1}\,K^{-1}}$, so the window spans 17% of the $8\,\mathrm{mS\,cm^{-1}}$ cut and a quarter of the formulations sit within 20% of it. Label temperature is moreover confounded with salt and with source, so the salt holdouts change the measurand as well as the chemistry. Two entries of the salt column, `LiTFSI` and `LiN(CF3SO2)2`, are the same compound under

two names, 27 and 9 room-temperature rows whose mean label temperatures differ by five degrees; they were encoded separately and can therefore be split across a salt holdout. Both should be repaired before the electrolyte results are relied on: the first by interpolating each formulation to a common temperature, the second by merging the names. The guarantees refer to the labels as recorded. Finally, we treat binary endpoints only, and one model throughout.

5 Conclusions

The coverage level of a conformal classifier is a coordinate for navigating screening decisions, and the audited consequences, counted in compounds and bounded with a stated probability, are what a team chooses by under its own objective and guarantees. Each setting sorts a future batch into one table of prediction set by true label, and every screening quantity is a sum of its cells. A labelled audit set from the library to be screened predicts all of these counts for the next batch, with intervals that hold with a specified probability in finite samples, under exchangeability alone and after a setting is chosen. The same request led to very different consequences across twelve Tox21 endpoints, and one model offered more than a hundred distinct trade-offs for a solubility screen and for an electrolyte screen. When the calibration did not match the screened population, as for a new lithium salt, the audit still bounded the consequences correctly. In automated campaigns, which run in parallel batches, the same construction applies to each batch, and the bounds become capacity constraints a planner can check. Inverted, the same intervals size a round: the batch that delivers a target number of hits with a stated probability is larger than dividing the target by the rate suggests, by a factor that grows as the rate falls, and the audit that pins the rate down is the measurement with the largest effect on what the round costs. What the audit must contain is decided by the batch it prices, since an audit that reproduces a batch's composition without sampling its chemistry prices nothing. With an audit built this way, a team sees those consequences before the campaign starts.

Author contributions

P.H.Z. conceived the study, developed the methodology, performed the analysis and wrote the manuscript.

Conflicts of interest

There are no conflicts to declare.

Data availability

Tox21 is available through MoleculeNet, AqSolDB through its dataset reference[49] and CALiSol-23 from the DTU data repository (DOI 10.11583/DTU.c.6929599).[50] The carbonic anhydrase selection counts and affinities are from the CA-DEL release[56] (Zenodo record 17656024, DOI 10.5281/zenodo.17656024, CC BY 4.0), whose affinity values derive from ChEMBL[57] and carry CC BY-SA 3.0. The DNA-encoded library is BELKA,[51] available from Leash Biosciences through the Polaris record `leash-bio/belka-v1` under CC BY-NC-SA 4.0; the million-molecule sample used here is reproduced by the seeded digest rule given in ESI Section S11. The scripts that reproduce every table and figure are provided with the ESI, together with the processed data, split indices and stored model scores. They implement the finite-sample threshold corrections, randomized tie-breaking, outcome tables, prediction intervals for future batches from an audit set or from the calibration set alone, coverage-request sweeps with corrections for selection, and the cost-ratio analysis. [Repository URL, version, license and archived DOI.]

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